

21 A chemical reaction has three reactants, X, Y and Z. The reaction is zero order with respect to [X], first order with respect to [Y] and second order with respect to [Z].

An experiment is performed in which $[X] = 0.0300 \text{ mol dm}^{-3}$, $[Y] = 0.0500 \text{ mol dm}^{-3}$ and $[Z] = 0.0400 \text{ mol dm}^{-3}$. The initial rate of reaction is $4.68 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$.

What is the numerical value of k in the rate equation?

A 0.234

B 5.85

C 7.80

D 117